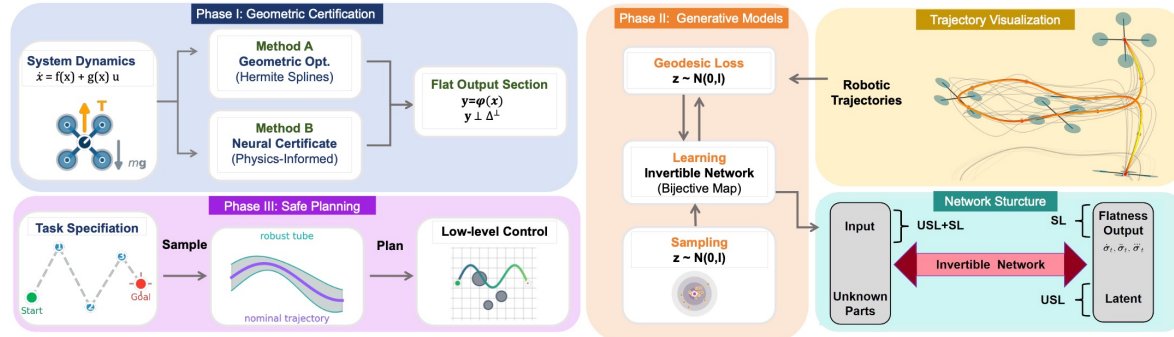


Neural Geometric Flatness: Certification, Trivialization, and Safe Planning

Pipeline overview



A unified neural pipeline for geometric flatness certification, stochastic mapping, and chance-constrained safe planning.

Part1 Geometric Flatness Certification

Constraint: Orthogonality to Underactuation Δ

$$T_s \sigma(T_s S) \perp \Delta_{\sigma(s)} \iff \text{proj}_{\Delta}(T_s \sigma \cdot v) = 0$$

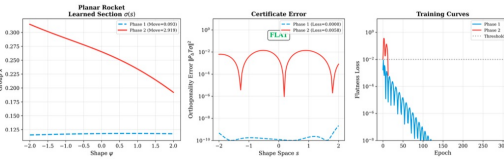
A Physics-Informed Neural Certificate (PINN) solves the variational flatness condition.

TABLE I: Flatness Certification Results: FEM vs. PINN

System	Manifold	Truth	FEM (Spline)		PINN (Two-Phase)	
			Loss	Verdict	Loss	Verdict
Planar Rocket	$SE(2)$	Flat	0.000	Trivial [†]	0.006	Correct
Aerial Manip.	$SE(2) \times S^1$	Flat	0.000	Trivial [†]	0.011	Correct
Quadrotor 2D	$SE(2)$	Flat	0.000	Trivial [†]	0.022	Correct
Acrobot	$S^1 \times S^1$	Not Flat	0.000	Wrong	0.000*	Correct
Cart-Pole	$\mathbb{R} \times S^1$	Not Flat	0.002	Correct	1.000	Correct

[†]FEM converges to $w = 0$ (constant section) for all systems.

*Phase 1 loss; Phase 2 loss = 0.332, indicating trivial-only solution.



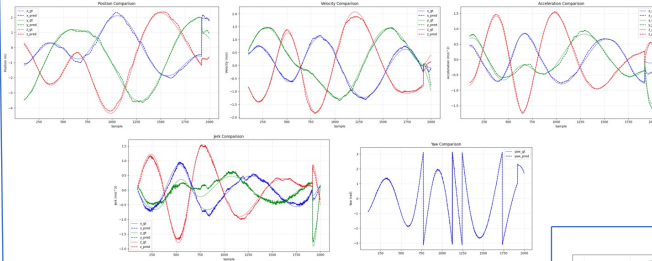
(a) Planar Rocket (Flat)

Part2 Stochastic INN Flatness Learning

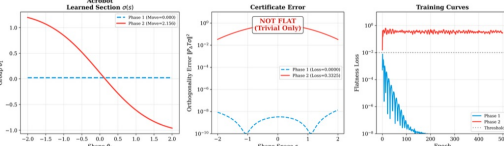
Stochastic Mapping:

$$(x, u) = f_{\theta}(\mathbf{y}_{jet}, z), \quad z \sim \mathcal{N}(0, I)$$

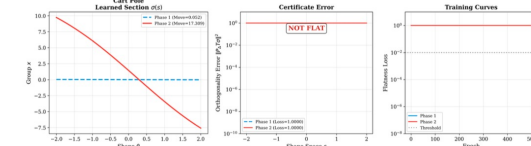
Maps flat outputs and latent noise z to full states, capturing real-world uncertainty.



(b) Quadrotor 2D (Flat)



(c) Acrobot (Trivial Only)



(d) Cart-Pole (Geometric Obstruction)

Part3 Safe Trajectory Planning

$$\min_c \int_0^T \|\mathbf{y}^{(4)}(t)\|^2 dt \quad (\text{Minimum Snap})$$

$$\text{s.t. } \mathbb{P}_z(h_{\text{safe}}(f_{\theta}(\mathbf{y}(t), z)) \leq 0) \geq 1 - \epsilon$$

Ensures thrust, tilt, and acceleration constraints are satisfied.

$$\mathcal{C}_{\text{safe}} = \{(x, u) \mid \|a\| \leq a_{\text{max}}, \phi \leq \phi_{\text{max}}, 0 \leq T \leq T_{\text{max}}\}$$

Outperforms baseline MINSNAP and heuristic scaling.

TABLE VI: Violation ratios and timing for the representative trajectory. "Dur." denotes normalized execution time (Baseline reference = 900).

Planner	Scale	Dur.	$v > v_{\text{max}}$	$\ a_{\text{cmd}}\ > a_{\text{max}}$	$\ a_{\text{cmd},xy}\ > a_{\text{max}}$	$\ \omega\ > \omega_{\text{max}}$
Baseline MINSNAP	0.90	900	11.8%	10.2%	7.4%	10.1%
Heuristic safe MINSNAP	1.14	1142	0.00%	0.00%	0.00%	0.00%
Learned (NN majority)	1.01	1010	0.00%	0.00%	0.00%	0.00%

$$\min_{c, \tau} \int_0^T \|j^{\kappa} \gamma(t)\|_Y^2 dt + w_{\tau} f(\tau),$$

$$\text{s.t. } j^{\kappa-1} \gamma(0) = \mathbf{y}_{\text{start}}, \quad j^{\kappa-1} \gamma(T) = \mathbf{y}_{\text{goal}},$$

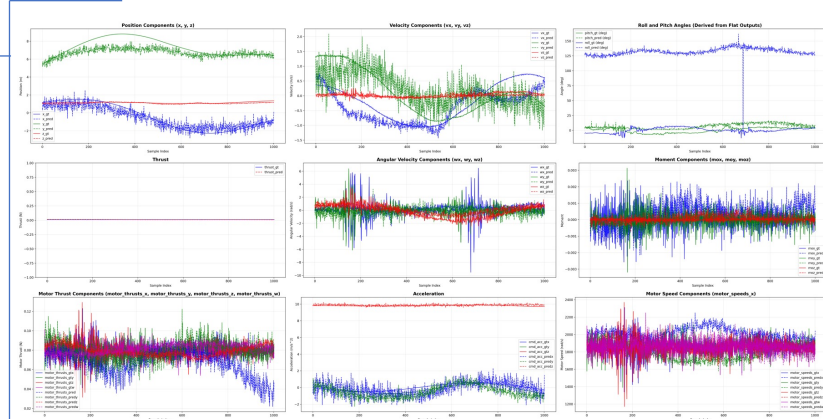
$$j^{\kappa-1} \gamma_i(\Delta t_i) = j^{\kappa-1} \gamma_{i+1}(0),$$

$$j^{\kappa-1} \gamma(t) \in \mathcal{Y}_{\text{adm}},$$

$$\forall i \in \{1, \dots, M\}, \quad \forall t \in [0, \Delta t_i].$$



Chance-constrained flat-output trajectory optimization.



INN generalizes to real hardware and accurately predicts trajectory deviations